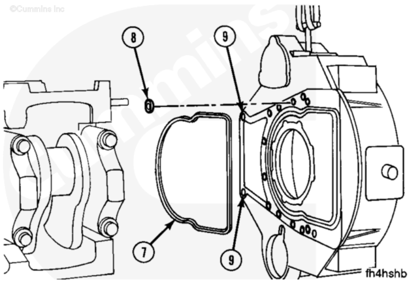
Trainings

Install

Note : Be sure the flywheel housing dowels have been installed in the block.

Install the sealing ring (7) in the groove on the housing. If a wet type is used, install the seals (8) in the counterbores as shown. The holes (9) do **not** require seals.

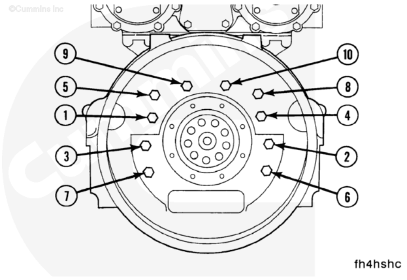
Use guide bolts to help during alignment. Install the housing and the capscrews.



Tighten the flywheel housing capscrews using the sequence shown.

Torque Value:

1. 100 n•m [75 ft-lb]
2. 205 n•m [150 ft-lb]

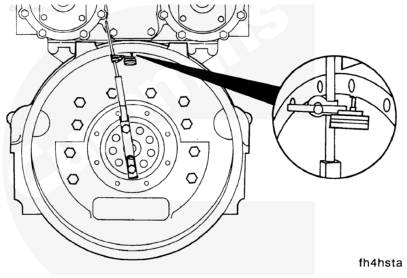


Measure the flywheel housing alignment.

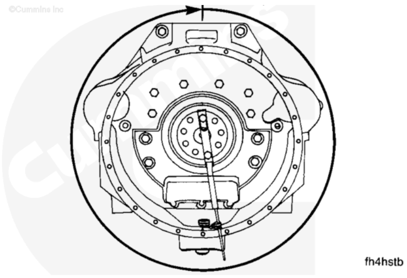
Note : The bore and the face of the housing must be in alignment with the crankshaft.

Note : The indicator arm must be rigid for an accurate reading. It must not sag.

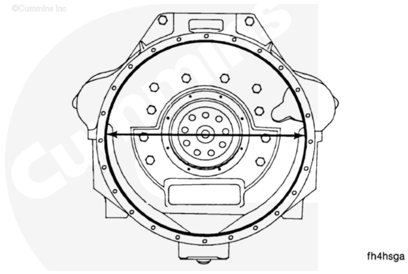
Attach an indicator to the crankshaft as shown.



Position the indicator at the 12 o'clock position. Adjust the dial until the needle points to ZERO. Rotate the crankshaft one complete revolution (360 degrees). Record the top indicator reading (TIR).



The maximum allowable TIR depends on the diameter of the bore.



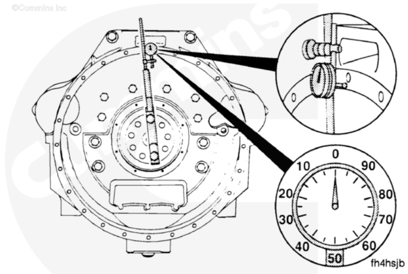
Bore Diameter/Maximum Top Indicator Reading (TIR)				
Minimum to Maximum mm	Minimum to Maximum [in]	SAE	mm	[in]
787.4 to 810.5	[31.00 to 31.91]	00	0.30	[0.012]
647.7 to 648.0	[25.50 to 25.51]	0	0.25	[0.010]
584.2 to 584.4	[23.00 to 23.008]	1/2	0.25	[0.010]
511.2 to 511.3	[20.125 to 20.13]	1	0.20	[0.008]

If the alignment is **not** within specifications and the bore is round, the housing can be shifted.

If the alignment is **not** within specifications and the bore is **not** round, the housing **must** be replaced.

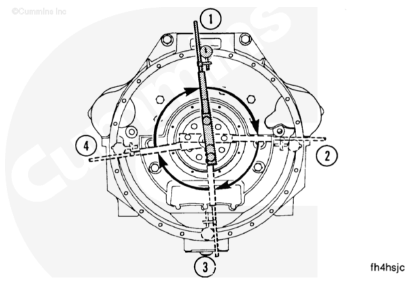
Note : The crankshaft end clearance must be pushed or pulled in the same direction each time a point is measured.

Attach an indicator as shown. Position the indicator at the 12 o'clock position. Adjust the dial until the needle points to ZERO.



Record the indicator reading at three different points; 3 o'clock, 6 o'clock, and 9 o'clock.

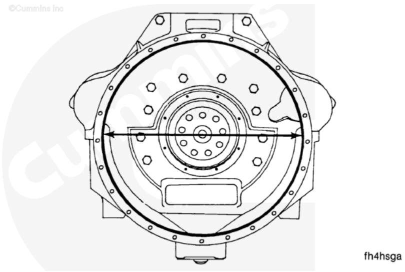
Turn backward to the original position. Be sure the needle still points to ZERO. Determine the total indicator runout (TIR).



		mm	[in]
Example:	3 o'clock	0.00	[0.00]
	6 o'clock	+0.08	[+0.003]
	9 o'clock	-0.05	[-0.002]
Equal TIR:		0.13	[0.005]

The maximum allowable TIR depends on the diameter of the bore.

Bore Diameter/Maximum TIR				
Minimum to Maximum mm	Minimum to Maximum [in]	SAE	mm	[in]
787.4 to 810.5	[31.00 to 31.91]	00	0.30	[0.012]
647.7 to 648.0	[25.50 to 25.51]	0	0.25	[0.010]
584.2 to 584.4	[23.00 to 23.008]	1/2	0.25	[0.010]
511.2 to 511.3	[20.125 to 20.13]	1	0.20	[0.008]

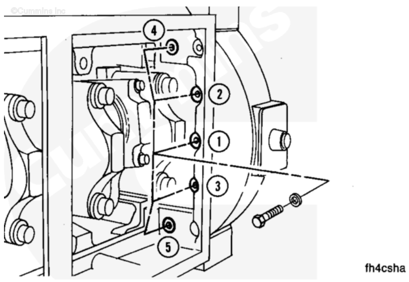


Note : If the alignment is not within specifications, remove the housing. Check for nicks, burrs, or foreign material between the block and the housing. Check the alignment again. If the alignment is not within specifications, the block or the housing is not machined correctly.

Tighten the five 9.5 mm [3/8-in] washers and capscrews in the sequence shown.

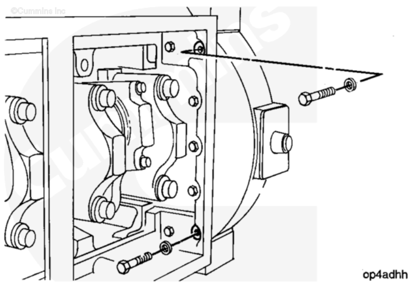
Torque Value:

1. 25 n•m [20 ft-lb]
2. 40 n•m [30 ft-lb]
3. 45 n•m [35 ft-lb]



Install the two 11 mm [7/16-in] washers and capscrews as shown. Tighten the capscrews.

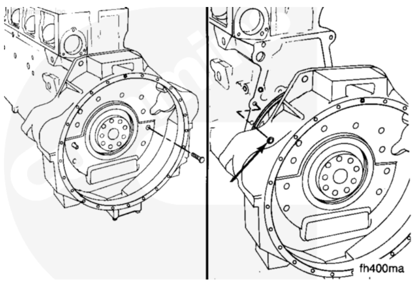
Torque Value: 65 n•m [48 ft-lb]



Install the flywheel housing mounting capscrews.

Tighten the capscrews.

Torque Value: 205 n•m [150 ft-lb]



Last Modified: 13-Feb-2026